

IN4MATX 285:

Interactive Technology Studio

Programming:
Design Systems

Today's goals

By the end of today, you should be able to...

- Articulate the goal of design systems
- Set up a website to access Adobe's Spectrum design system
- Implement components of a design system in a website

What is a Design System?

Why do we use them?

Caveat: my biases

Design Systems

- You work for a large company, and are tasked with designing the layout for a new app to add to their existing suite
- How do you start?
 - You probably look at the current apps and try to replicate the look and feel
- Why?
 - Consistency across products
 - Reduce the design choices you have to make

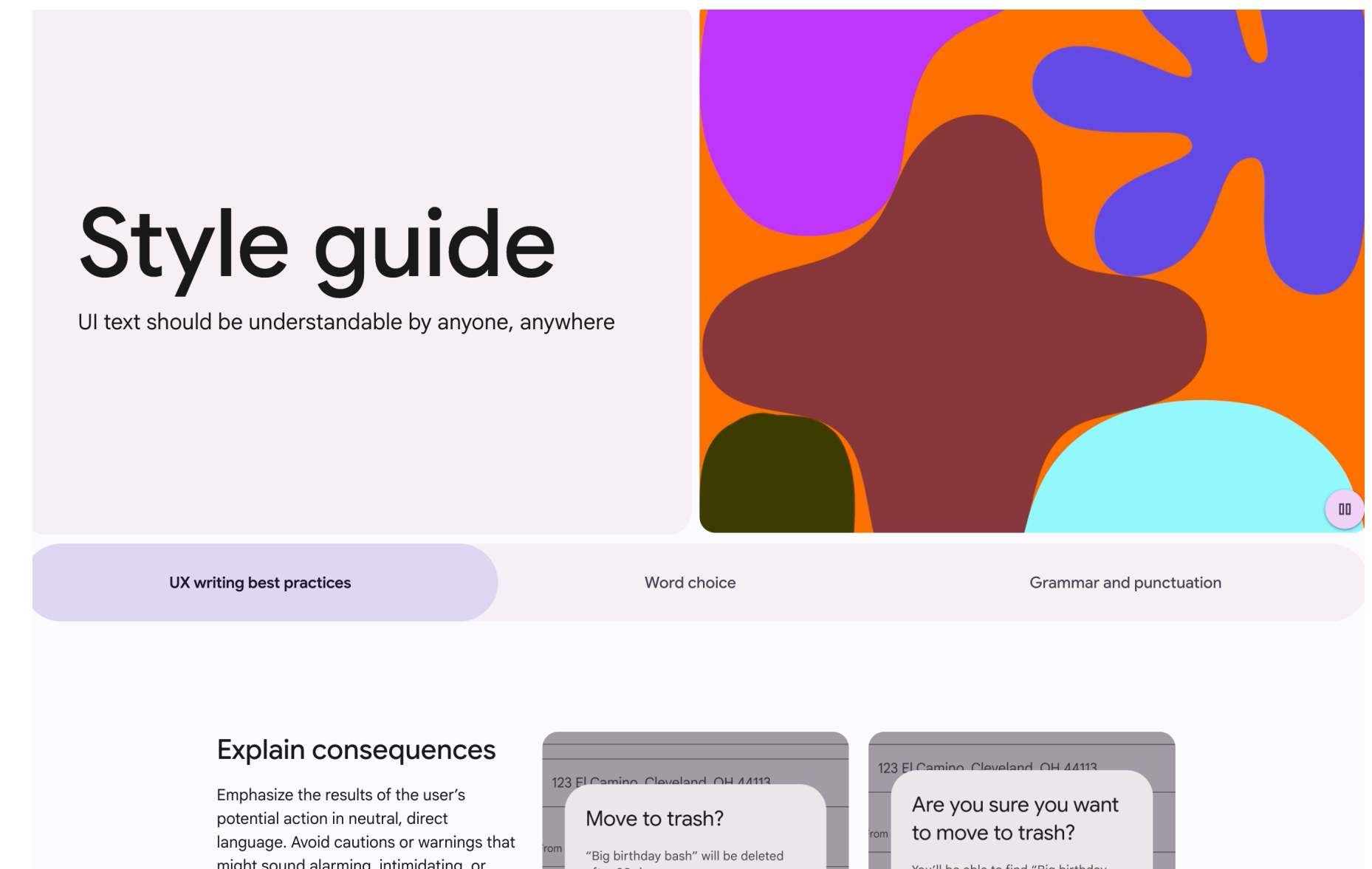
Design Systems

- Aim to *formalize* the UI choices made across software/systems
- Support *reusability* by offering guides and implementations
- Common components: style guide, component library, pattern library

Design Systems

Style guide

- Offers implementation guidelines & visual references
 - How should text be written?
 - Visual interaction standards
 - Sometimes branding



<https://m3.material.io/foundations/content-design/style-guide/ux-writing-best-practices>

Design Systems

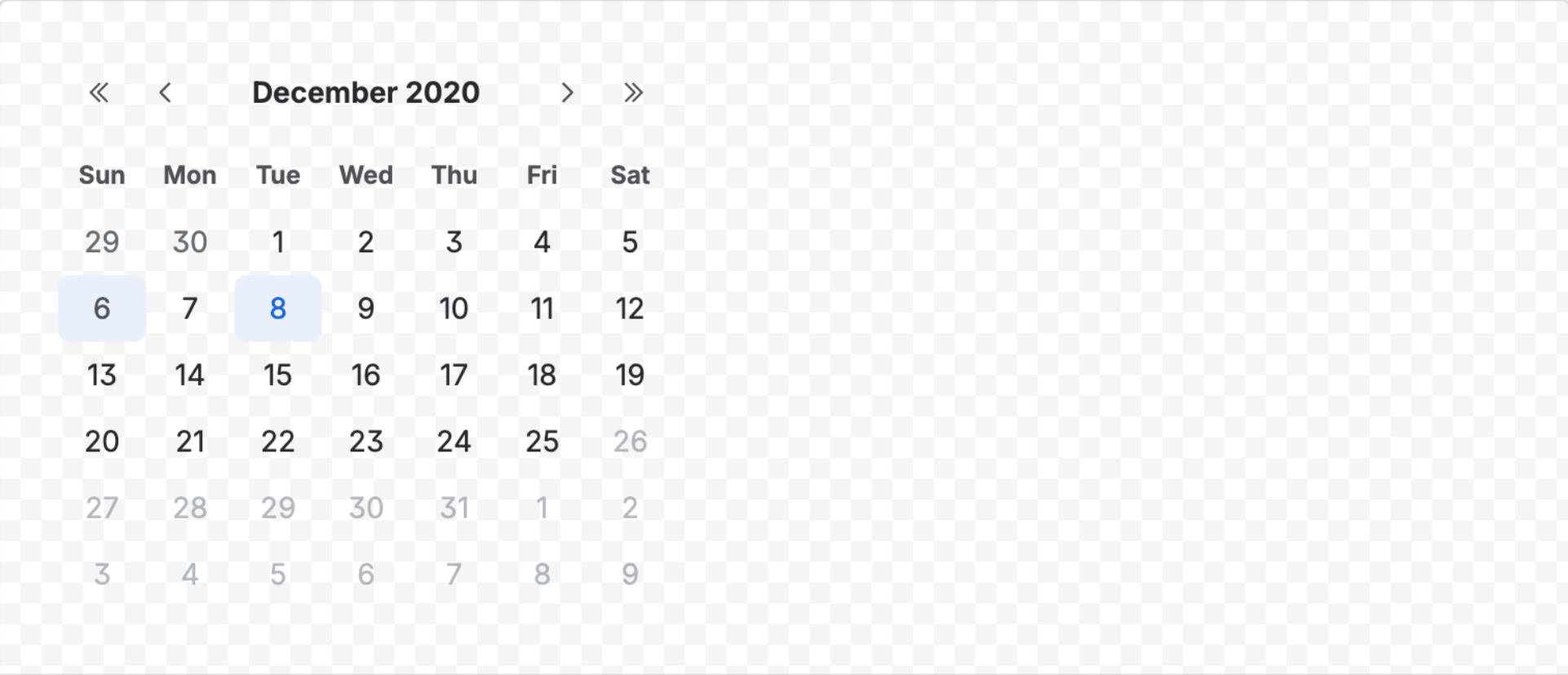
Component library

- Reusable UI elements
 - Named, described
 - Attributes and states
 - Often code snippets

<https://atlassian.design/components/calendar/examples>

Default

The calendar component provides a way to render dates for selection or presentation purposes.



The screenshot shows a calendar component for December 2020. It features a grid of days with the 8th highlighted in blue. Navigation arrows are present at the top. Below the calendar, there are controls for styles, a 'Copy code' button, and a link to 'Edit in code sandbox'.

```
import React from 'react';

import Calendar from '@atlassian/calendar';

const defaultPreviouslySelected = ['2020-12-06'];
const defaultSelected = ['2020-12-08'];

export default (): React.JSX.Element => (
  <Calendar
```

SHOW MORE ▾

Design Systems

Pattern library

- Collections of UI-element groupings designed for use together
- Worked examples of common use cases

🇺🇸 An official website of the United States government [Here's how you know](#) ▾

<Project title>

Create account

Get started with an account.

* indicates a required field.

Email address *

Create password

[Show password](#)

Re-type password *

☐ I agree to the [terms and conditions](#). *

Create account

Already have an account? [Sign in](#).

<A tagline that explains the benefit of creating an account.>

<Here's space for a longer description to introduce 3-5 easily scannable bullet points. Note that on mobile, this text block will bump below the Create Account form.>

- <Value proposition 1: Vivamus nec velit sed leo scelerisque laoreet vestibulum.>
- <Value proposition 1: Vivamus nec velit sed leo scelerisque laoreet vestibulum.>
- <Value proposition 3: Vivamus nec velit sed leo scelerisque laoreet vestibulum.>

Are you a federal employee?

If you are a federal employee or [other secondary user], please use [secondary Single Sign On (SSO)].

Launch secondary SSO

[Return to top](#)

<Primary link> <Primary link> <Primary link> <Primary link> <Primary link>

 <Name of Agency>



<Agency Contact Center>
<(800) 555-GOVT> <info@agency.gov>

<https://designsystem.digital.gov/templates/authentication-pages/create-account/>

**So how do we code with Design
Systems?**

Design Systems

- Many Design Systems are *open source*
 - All materials are publicly available and licensed for commercial use
- These open source systems provide an implementation you can extend

Design Systems

- Some examples
 - Material (Google)
 - Spectrum (Adobe)
 - Fluent (Microsoft)
 - Carbon (IBM)
 - Atlassian Design System
 - USWDS (U.S. Government)

Coding with one example:

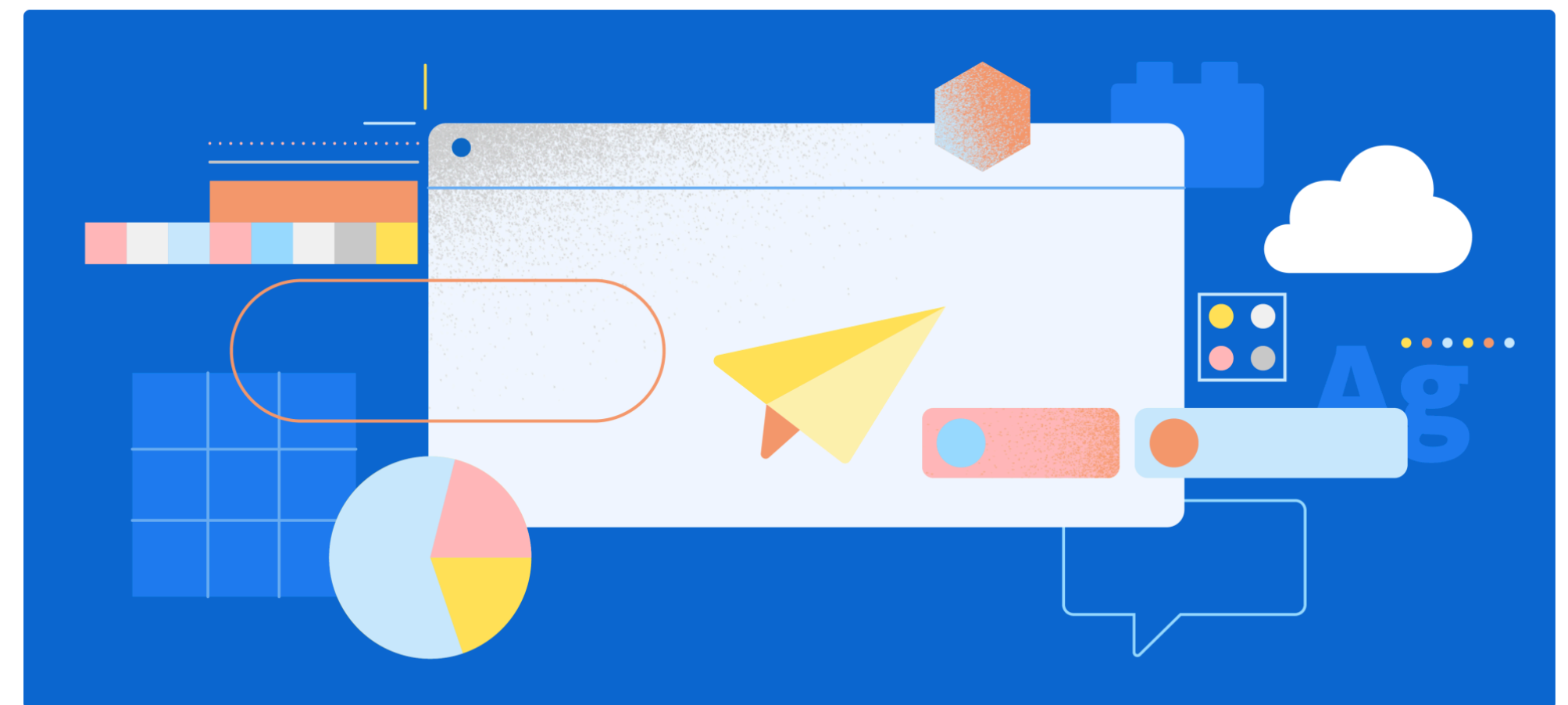
Adobe Spectrum

Adobe Spectrum

- Includes a rich component library capable of customization
- Offers Figma UI Kit for mockups
- Their Web Components don't require any imports or other software knowledge
 - Most build on React or require web servers, which are beyond the scope of this class

Spectrum, Adobe's design system

Spectrum provides components and tools to help product teams work more efficiently, and to make Adobe's applications more cohesive.



<https://opensource.adobe.com/spectrum-web-components/>

Adobe Spectrum

- Import it, and you can use its components

`<head>`

```
  <script src="https://jspm.dev/@spectrum-web-components/  
bundle/elements.js" type="module" async></script>
```

`</head>`

- On web: typically dedicated tags and CSS attributes

Adobe Spectrum

- New custom tag
- The system adds CSS attributes to render as expected

```
<sp-textfield id="address"
name="address" type="text"
placeholder="Address">
</sp-textfield>
```

Address

```
:host {                                     constructed stylesheet
  display: inline-flex;
  flex-direction: column;
  inline-size: var(--mod-textfield-width, var(
    --spectrum-textfield-width));
}

:host {                                     constructed stylesheet
  --spectrum-textfield-border-color:
    var(--system-spectrum-textfield-border-color);
  --spectrum-textfield-border-color-hover:
    var(--system-spectrum-textfield-border-color-ho);
  --spectrum-textfield-border-color-focus:
    var(--system-spectrum-textfield-border-color-fc);
  --spectrum-textfield-border-color-focus-hover:
    var(--system-spectrum-textfield-border-color-fc);
  --spectrum-textfield-border-color-keyboard-focus:
    var(--system-spectrum-textfield-border-color-ke);
  --spectrum-textfield-border-width: var(
    --system-spectrum-textfield-border-width);
}
```


Adobe Spectrum

- Include supported customization options
 - For example, what CSS attributes are supported, and what they do

Attributes and Properties				
Property	Attribute	Type	Default	Description
autocomplete	autocomplete	'list' 'none' HTMLInputElement['autocomplete'] HTMLTextAreaElement['autocomplete'] undefined		What form of assistance should be provided when attempting to supply a value to the form control
disabled	disabled	boolean	false	Disable this control. It will not receive focus or events
grows	grows	boolean	false	Whether a form control delivered with the `multiline` attribute will change size vertically to accomodate longer input
invalid	invalid	boolean	false	Whether the `value` held by the form control is invalid.
label	label	string	' '	A string applied via `aria-label` to the form control when a user visible label is not provided.
maxlength	maxlength	number	-1	Defines the maximum string length that the user can enter
minlength	minlength	number	-1	Defines the minimum string length that the user can enter

Implementing Spectrum



My 285 Schedule

Monday	Tuesday	Wednesday	Thursday	Friday
Discussion	Lecture		Lecture	
Discussion Assignment due	Lecture		Lecture	

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Implementing Spectrum



Who are we talking to?

Name:

Say hi

Closing thoughts on Design Systems

- Documentation is crucial to design systems
 - For creators who want their system to get adopted
 - For UX designers who want to understand how to use a system

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